INFORMATION PAPER

Network Throughput Optimization: Results and Recommendations

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Date: October 23, 2026
Subject: Technical assessment submission: latency-aware routing of four traffic streams
Attachments: Executive deck (.pptx); analysis notebook (.ipynb); notebook export (PDF)

1. Purpose

This paper summarizes the findings of the network throughput optimization assessment and the recommendations that follow from them. It is written to be read on its own. The attached notebook contains the full formulation, code, validation checks and supporting analysis, and the deck presents the same results in briefing form.

Bottom line

The network can deliver at most **83 of the 126 Mbps** offered at the snapshot studied (65.9%). This ceiling is physical, not a routing failure. Three directed links, **B→D, C→UK and C→US**, form the network's minimum cut, and their capacities sum to exactly 30 + 25 + 28 = 83 Mbps. They are the only links where an extra 1 Mbps raises delivered throughput.

Routing each stream on its own shortest path is infeasible: it would load the 18 Mbps B→C link with 61 Mbps (339%). Single-path routing reaches the same 83 Mbps as splittable routing in every case tested, so it is sufficient for throughput. Splitting is still worth keeping, because it lowers latency for sensitive traffic. Finally, maximizing throughput alone leaves 7 of 20 source-stream flows with no service, so a fairness or SLA policy should be an explicit step in any production optimizer.

So what: better routing is a free and immediate gain in feasibility and call quality. Serving more customers requires targeted capacity at three links, and protecting every customer requires an explicit fairness policy.

Answers to the assessment questions

Assessment question	Answer	See
How much of each stream's demand is delivered?	Split: UK-L 16, UK-T 34, US-L 18, US-T 15 Mbps. Single-path: 12, 36, 18, 17 Mbps.	Table 2
Which path, or split across paths, does each stream use?	Sensitive streams use one fast path per source through C. Tolerant streams use the D corridor and are the only ones that split. 18 split / 15 single-path routes.	4.3, App. E
How much of each stream cannot be delivered?	Split: 10, 5, 11, 17 Mbps. Single-path: 14, 3, 11, 15 Mbps (UK-L, UK-T, US-L, US-T).	Table 2

Assessment question	Answer	See
What is the total delivered across all four streams?	83 of 126 Mbps (65.9%) in both variants, equal to the network's minimum cut.	4.2
Which links are bottlenecks, and what does one more unit of bandwidth add?	B→D, C→UK and C→US: +1 Mbps each (83 → 84). Every other link: 0.	4.6
How should throughput and latency be defined and balanced?	Lexicographically: maximize delivered Mbps, then minimize sensitive Mbps·ms with each admission locked.	3
How do the single-path and split variants compare?	Same 83 Mbps. Splitting lowers sensitive latency: UK-L 18.25 vs. 21.17 ms; US-L 19.56 vs. 24.78 ms.	4.4, 4.5
What happens when demand changes?	The model is re-solved per snapshot: at 60% demand all 75.6 Mbps is delivered; at 150% delivery stays at 83 Mbps.	4.7

Assessment crosswalk. Each required output of the problem statement, with its answer and the section that supports it.

2. Background

The network carries traffic to two exits, UK and US. Each exit receives two streams. A **latency-sensitive** stream (UK-L, US-L) behaves like a video call and loses value on a slow route. A **tolerant** stream (UK-T, US-T) behaves like a file transfer and only needs to arrive. All four streams share the same links, and every link has a fixed bandwidth in each direction and a fixed latency. If tolerant traffic fills the fastest links, sensitive traffic is pushed onto slower ones, so the four streams must be planned together rather than one at a time. The problem is solved for a single instant: demand is fixed, nothing queues, and demand that cannot be routed is lost.

The network has nine nodes and eleven full-duplex links. Five nodes generate traffic: the edge nodes S1, S2 and S3, and the inner nodes B and C. Every route to an exit must pass through C or D. Because no production figures were provided, I built a synthetic dataset with deliberately uneven values: link capacities range from 18 to 40 Mbps and latencies from 7 to 18 ms. This gives real tradeoffs between fast-but-narrow and slow-but-wide routes. Demand is specified per source node and stream, giving 20 commodities and 126 Mbps of total offered demand (UK-L 26, UK-T 39, US-L 29, US-T 32). Appendix A lists every link and demand value. All values are parameters, so real telemetry can replace them without changing the model. The model is also layout-agnostic: the simpler 8-node variant with a single hub in front of both exits, which the problem statement also covers, runs by swapping in its link table.

Before modeling, I sent six clarifying questions (Appendix C). The answers fixed the following: demand is known per source node; the single-path rule applies to each source-stream pair separately; a single-path commodity may be partially delivered; nodes have no capacity limit and do not hold traffic; and sensitive traffic receives no admission priority, because all delivered traffic is weighted equally. These answers shape the results described below.

3. Approach

I modeled the problem as a multi-commodity flow over 22 directed arcs and 68 enumerated loop-free candidate paths, and solved it in Python with PuLP and the CBC solver. The objective is **lexicographic** rather than weighted. Stage 1 maximizes total delivered throughput. Stage 2 then minimizes the latency exposure of sensitive traffic, measured in Mbps·ms (flow multiplied by path latency), while holding each

commodity's Stage 1 delivered amount fixed. I chose this structure for two reasons. First, it avoids inventing an exchange rate between megabits and milliseconds. Second, it closes a loophole: if only the total were held fixed, Stage 2 could reduce sensitive latency simply by dropping sensitive traffic and admitting the same amount of tolerant traffic.

Both required routing variants were solved. The **splittable** variant lets a commodity use any mix of paths and is a linear program with 68 variables. The **single-path** variant adds one binary selection variable per path, limits each commodity to at most one selected path, and allows partial delivery on that path. It is a mixed-integer program with 136 variables. I then compared the results against a naive baseline in which each commodity independently takes its lowest-latency path. Bottlenecks were identified by adding 1 Mbps to each of the 22 arcs in turn and re-solving. Finally, the models were stress-tested under five demand and capacity scenarios.

4. Findings

4.1 Independent shortest-path routing is infeasible

When each commodity takes its own fastest route, nearly all traffic converges on node C. The slower D corridor (B→D, D→UK, D→US) carries nothing. Four links are overloaded, as shown below. The broader lesson is that sensible local choices can combine into a plan that cannot be executed when they share resources. A capacity-aware optimizer is therefore required before latency or fairness can even be discussed. **So what:** without central optimization, the network would congest its fastest corridor while leaving the D corridor idle, losing traffic it actually has room to carry.

Directed link	Capacity (Mbps)	Naive load (Mbps)	Utilization
B→C	18	61	339%
C→US	25	61	244%
C→UK	30	65	217%
A→C	25	45	180%

Table 1. Links overloaded under naive minimum-latency routing.

4.2 Maximum throughput is 83 Mbps, a hard physical ceiling

Optimal routing delivers 83 Mbps and loses 43 Mbps. It does this mainly by moving traffic off the C corridor and onto the D corridor: B→D rises from 0 to 28 Mbps, D→UK from 0 to 20, and D→US from 0 to 8. The ceiling holds even when demand rises to 150%. The reason is structural. If S1, S2, S3, A, B and C are placed on one side and D, UK and US on the other, the only arcs that cross are C→UK (30), C→US (25) and B→D (28). Their total, 83 Mbps, is the minimum cut, and by the max-flow/min-cut theorem no routing can exceed it. Delivering more therefore requires added capacity, reduced demand, or a change in admission policy. Better routing cannot do it. **So what:** 83 Mbps is the baseline capacity KPI. A third of today's demand, and more than half at 150% demand, cannot be served, so revenue from traffic growth requires capacity investment, not better software.

Stream	Offered	Split: delivered	Split: lost	Single: delivered	Single: lost
UK-L	26	16	10	12	14
UK-T	39	34	5	36	3

Stream	Offered	Split: delivered	Split: lost	Single: delivered	Single: lost
US-L	29	18	11	18	11
US-T	32	15	17	17	15
Total	126	83	43	83	43

Table 2. Delivered and undelivered demand by stream, final solutions (Mbps). Both models deliver 83 Mbps in total but split it differently across streams.

4.3 Where each stream's traffic travels

Table 3 summarizes the routes each stream uses in the final solutions. Appendix E lists all 18 splittable and 15 single-path routes with their flow and latency. In both models, sensitive traffic is kept on the short routes through C. In the splittable solution, UK-L uses S1→A→C→UK (27 ms), B→C→UK (17 ms) and C→UK (7 ms), and US-L uses S2→A→C→US (31 ms), B→C→US (19 ms) and C→US (9 ms). Each sensitive commodity uses a single path. Splitting occurs only in the tolerant streams, where five commodities divide their traffic across two paths. This frees the fast links for sensitive traffic, and the tolerant traffic absorbs the slower D corridor.

Two features of the routes deserve attention. First, some tolerant routes are long, up to 66 ms (S2→A→C→B→D→UK), because Stage 2 does not price tolerant latency. They are valid but untidy, and the cleanup step recommended in Section 6 would shorten them without changing any earlier result. Second, under single-path routing, source B's US-L traffic is assigned to the slower B→D→US route (32 ms). This one choice is the main reason single-path US-L latency is higher. Single-path partial delivery is also visible: C's UK-L commodity delivers 1 of 4 Mbps on its one route. **So what:** the routing is explainable to an operations team: sensitive traffic takes the fast C corridor and tolerant traffic gives way, which is the behavior the business wants.

Stream	Splittable routes (Mbps)	Single-path routes (Mbps)
UK-L	S1→A→C→UK 6 · B→C→UK 6 · C→UK 4	S2→A→C→UK 5 · B→C→UK 6 · C→UK 1
UK-T	9 routes; largest B→D→UK 9.5, S2→A→B→D→UK 5.5, B→A→C→UK 4.5	S1→A→B→D→UK 4 · S2→A→B→C→UK 6 · S3→B→A→C→UK 7 · B→D→UK 14 · C→UK 5
US-L	S2→A→C→US 5 · B→C→US 8 · C→US 5	S1→A→C→US 5 · B→D→US 8 · C→US 5
US-T	B→A→C→US 7 · B→D→US 2 · C→A→B→D→US 6	S1→A→B→C→US 6 · S3→B→A→C→US 3 · B→D→US 2 · C→US 6

Table 3. Routes used by each stream in the final solutions. Full route list in Appendix E.

4.4 Single-path routing matches splittable routing on throughput

In theory, single-path throughput can never exceed splittable throughput. In this network the two are equal: 83 Mbps at baseline and identical in all five stress scenarios. Even the splittable optimum uses only one path per commodity at Stage 1. The single-path restriction therefore costs nothing in throughput, which favors the simpler routing scheme operationally. Equal totals do not mean equal solutions, however. The two models serve 10 of the 20 commodities differently; for example, UK-L receives 16 Mbps under splitting and 12 Mbps under single-path. Many different admission patterns reach 83 Mbps, so the solver's choice among them has real service consequences. **So what:** the operationally simpler design can be adopted with no loss of delivered traffic.

4.5 Splitting earns its value in latency

Stage 2 shows what routing flexibility is worth once throughput is fixed. In the splittable model, re-routing the admitted traffic cut sensitive latency exposure from 850 to 644 Mbps·ms, a 24.2% reduction, without losing any throughput. Average sensitive latency in the final solutions is also lower with splitting, as Table 4 shows. One caveat applies: the two pipelines admit different traffic mixes, so this gap combines a routing effect with an admission-mix effect. It is a fair comparison of the two complete pipelines, but not a controlled measure of splitting alone. **So what:** customers on calls get a measurably faster experience from a software change alone, with no capital spending.

Measure	Splittable	Single-path
Maximum throughput	83 Mbps	83 Mbps
UK-L average / worst-case latency	18.25 / 27 ms	21.17 / 29 ms
US-L average / worst-case latency	19.56 / 31 ms	24.78 / 32 ms
Saturated links in final solution	6	4
Model type and size	LP, 68 variables	MILP, 136 variables

Table 4. Final splittable vs. single-path solutions.

4.6 Saturation is not the same as a bottleneck

Six directed links reach 100% utilization in at least one solution. Only three of them add throughput when widened: an extra 1 Mbps on B→D, C→UK or C→US raises the optimum from 83 to 84 Mbps in both models. B→C, A→C and A→B are also fully used at times, but widening them adds nothing, because the traffic they carry must still cross one of the three cut links. A utilization dashboard would flag all six links. Only the marginal-value test identifies the three where investment converts into delivered service. This value is local: adding 10 Mbps to one link will not add 10 Mbps of throughput once another cut becomes binding. **So what:** an upgrade plan driven by utilization would put half its spend on links that return nothing; targeting the three cut links returns one delivered Mbps for each Mbps added.

4.7 Exit-facing capacity is the most fragile

Scenario	Offered (Mbps)	Max throughput (Mbps)	Delivery rate
Baseline	126.0	83.00	65.9%
Low demand (60%)	75.6	75.60	100.0%
High demand (150%)	189.0	83.00	43.9%
Internal corridor at 70%	126.0	78.10	62.0%
Fragmented internal capacity	126.0	72.85	57.8%
Exit links at 75%	126.0	69.25	55.0%

Table 5. Stress scenarios. Splittable and single-path results were identical in every case.

Degrading the exit-facing links costs the most throughput. The result, 69.25 Mbps, equals the same cut evaluated at reduced capacity (22.5 + 18.75 + 28), which confirms that capacity near the destinations governs service in this network. Higher demand is simply lost: at 189 Mbps offered, the network still delivers 83. **So what:** exit links are where redundancy and monitoring investment should go first. A 25%

exit loss costs 13.75 Mbps, nearly three times the 4.9 Mbps lost when the internal corridors degrade by 30%.

4.8 Maximum throughput does not guarantee fair service

Because every delivered Mbps counts equally, the optimizer favors traffic that enters close to the exits, since that traffic uses the least scarce capacity. In the splittable solution, sources B and C receive 100% of their demand, while S3 receives 2 of 24 Mbps (8%). Seven of the 20 commodities receive nothing, and five do under single-path. This is not a modeling error; it follows directly from the stated objective. It does show that a single throughput figure can hide very different customer outcomes. How scarce capacity should be shared is a business decision, and the optimizer should be told the answer explicitly. **So what:** without a stated policy, service losses fall on whichever sites are hardest to reach, which is where SLA breaches and customer churn would appear.

5. Business implications: the so what

Table 6 translates each finding into its business consequence and the action it supports. The paragraphs that follow draw out the four implications that matter most for leadership.

Finding	So what	Business impact	Action
Naive routing is infeasible	Local route choices collide on shared links	C corridor at up to 3.4× capacity while the D corridor sits idle	Run a joint, capacity-aware optimizer
83 Mbps hard ceiling	Routing cannot create capacity	34% of demand unserved today, 56% at +50% growth	Plan capacity and demand policy against the 83 Mbps KPI
Only 3 links have marginal value	Utilization dashboards mislead	3 of 6 saturated links return 0 Mbps if upgraded; cut links return 1:1	Direct capex to B→D, C→UK, C→US
Stage 2 and splitting cut latency	Quality improves without new capacity	−24% sensitive latency exposure; UK-L/US-L −14%/−21%	Keep multipath for latency-critical classes
Single-path matches throughput	Simplicity costs nothing at baseline	15 vs. 18 routes; deterministic paths	Default to single-path routing
7 of 20 flows unserved	Max throughput is not fair service	Losses concentrate on specific sites (S3 at 8%)	Add a fairness / SLA step
Exit degradation hurts most	Exits are the fragile point	25% exit loss costs 13.75 Mbps, nearly 3× internal	Prioritize exit redundancy

Table 6. From findings to business decisions.

Routing is necessary but not sufficient. A joint optimizer turns an infeasible plan into the best plan the physical network allows, and 83 Mbps becomes the baseline capacity KPI: any operating policy that delivers less is leaving feasible service unused. The remaining 43 Mbps shortfall is a capacity problem, and demand growth will be lost unless capacity, admission policy or demand shaping changes.

Investment should follow marginal value, not utilization. The three cut links are where the next unit of bandwidth produces service. Upgrades should be sized in steps, re-optimizing after each one, because once a cut link is relieved a different set of links becomes the constraint. Upgrade cost, resilience and expected demand growth should be weighed alongside the marginal value.

Operational simplicity is available at no throughput cost. Single-path routing delivers the same throughput with fewer active routes and simpler monitoring and troubleshooting. Multipath capability is worth retaining for latency-critical traffic and for degraded conditions.

Service reporting must go below the total. Delivery and latency should be reported by source and traffic class, because equal totals can conceal starved customers.

6. Recommendations

I recommend an adaptive, policy-aware optimizer that makes decisions in four ordered steps:

1. **Throughput:** maximize delivered Mbps subject to capacity, exit and demand rules.
2. **Fairness and SLA policy:** apply minimum delivery ratios, max-min fairness or class floors chosen by the business. This is the step the current model lacks.
3. **Sensitive latency:** minimize UK-L and US-L latency exposure with admissions locked.
4. **Operational cleanup:** shorten tolerant-traffic routes and reduce the number of active paths without changing the results of the earlier steps.

Under normal load, use single-path routing. Enable multipath for latency-critical classes or when the network is degraded. Prioritize capacity reviews for B→D, C→UK and C→US, and replace utilization-only dashboards with marginal-value analysis.

7. Limitations and next steps

These results are rigorous within the model's assumptions, and those assumptions define what the numbers mean. The data is synthetic, so 83 Mbps applies only to this network, although the method transfers. Latency is fixed per link, so modeled latency near full utilization is optimistic. Marginal value is a local, one-link-at-a-time measure, and joint upgrades were not tested. The optimizer is exact over the 68 enumerated paths; larger networks would need path generation. The model is a deterministic snapshot with no failure probabilities and no cost for changing routes.

In priority order, the next steps are: (1) add fairness and SLA constraints; (2) calibrate the model with real capacity, latency and demand telemetry; (3) add utilization-dependent latency; (4) build a throughput–latency frontier by capping sensitive latency at several levels and re-solving, so that leaders can see the tradeoff directly rather than having it fixed by rule; and (5) extend to larger networks and failure scenarios using column generation and robust optimization. A controlled latency experiment, with both models fixed to a common admission vector, would also isolate the value of splitting itself.

8. Conclusion

This assessment set out to route four competing traffic streams through a shared network so that as much demand as possible reaches its exit, while keeping latency-sensitive traffic fast. The answer has three parts. First, coordinated routing is essential: routing each stream independently produces a plan that cannot run, while joint optimization reaches the best result the network allows. Second, that best result is 83 of 126 Mbps, and it is fixed by physics rather than by routing. Three links, B→D, C→UK and C→US, form the network's minimum cut, and they are the only places where added bandwidth increases delivered service. Third, once throughput is secured, the remaining choices are about service

quality and policy. Single-path routing delivers the same throughput more simply. Multipath routing lowers latency for sensitive traffic. An explicit fairness rule is needed so that the drive for total throughput does not leave some customers unserved.

For leadership, the practical message is to treat network performance as a sequence of decisions rather than a single metric: establish what the network can physically carry, decide how scarce capacity should be shared, route the admitted traffic for the best service, and invest where the next unit of capacity actually converts into delivered traffic. The model built for this assessment supports each of those decisions today on synthetic data. With real telemetry and the fairness extension, it is ready to be developed into an operational planning tool.

Appendix A. Input data

Link	Capacity (Mbps, each direction)	Latency (ms)	Role
S1–A	40	12	Entry
S2–A	35	14	Entry
S3–B	30	10	Entry
A–B	20	15	Internal
A–C	25	8	Internal, fast
B–C	18	10	Internal, narrow
B–D	28	18	Internal, wide/slow · cut link
C–UK	30	7	Exit, fast · cut link
C–US	25	9	Exit, fast · cut link
D–UK	22	16	Exit, slow
D–US	30	14	Exit, slow

Table A1. Synthetic link dataset (11 full-duplex links, 22 directed arcs).

Source	UK-L	UK-T	US-L	US-T	Total
S1	6	7	5	6	24
S2	5	6	5	5	21
S3	5	7	6	6	24
B	6	14	8	9	37
C	4	5	5	6	20
Total	26	39	29	32	126

Table A2. Offered demand by source and stream (Mbps).

Appendix B. Frequently asked questions

Is 83 Mbps a solver artifact? No. It equals the minimum cut (C→UK + C→US + B→D = 83). Both models reach this bound, and every solve reported an optimal status.

Why not combine throughput and latency into one weighted objective? A weight would set an arbitrary exchange rate between Mbps and milliseconds and change the answer depending on that choice. The lexicographic order makes the priority explicit. A throughput–latency frontier is the recommended way to show leaders the tradeoff.

Why lock each commodity's admission in Stage 2 rather than only the total? Locking only the total would let Stage 2 lower sensitive latency by swapping sensitive traffic for tolerant traffic of the same volume. That would look like an improvement while actually delivering less of the traffic the stage is meant to protect.

If splitting adds no throughput, why keep it? It lowers sensitive latency (24.2% lower exposure within the splittable pipeline). Its value may also be greater on other topologies or under failures not tested here.

Why is S3 nearly unserved? S3's traffic must cross the most contested interior links to reach either exit, while B and C sit next to the exits. With every Mbps valued equally, serving B and C first is the cheapest way to reach 83 Mbps. A fairness constraint would change this.

Appendix C. Clarifying questions and answers

Before modeling, I sent six clarifying questions on October 16. The answers below define the final formulation.

Question asked	Answer	Effect on the model
Will demand be given per source node, or only per stream?	Per source node	20 source-stream commodities instead of 4 pooled streams
Does the single-path rule apply per stream or per source within a stream?	Per source-stream pair	Each source's share of a stream picks its own single path
If one path cannot carry all of a commodity, is partial delivery allowed?	Yes, partial delivery is allowed	$f \le D \cdot y$ rather than all-or-nothing admission
How should latency be measured when a sensitive stream is split?	Flow-weighted average; worst case also reported	Stage 2 minimizes Mbps·ms; worst-case path latency reported
Are 30 Mbps / 20 ms the values for every link, and will data be provided?	Illustrative only; build a representative dataset	Synthetic, uneven dataset; all values parameterized
Can nodes queue or hold traffic? Do nodes have capacity? Does sensitive traffic go first?	No queuing or carry-over; nodes unconstrained; no admission priority	Snapshot model; only links constrain; all Mbps weighted equally

Table C1. Clarification questions and how each answer shaped the formulation.

Appendix D. Model summary

Stage 1 maximizes $\Sigma\, z(s,k)$, where $z(s,k) = \Sigma\, f(p)$ over the candidate paths p of commodity (s,k). Constraints: $0 \le z(s,k) \le D(s,k)$; for every directed arc e, $\Sigma\, f(p)$ over paths using $e \le C(e)$; and paths end only at the commodity's own exit. The single-path variant adds binary $y(p)$ with $f(p) \le D(s,k) \cdot y(p)$ and $\Sigma\, y(p) \le 1$ for each commodity. Stage 2 minimizes $\Sigma\, L(p) \cdot f(p)$ over UK-L and US-L paths, where $L(p)$ is the sum of link latencies, subject to the same constraints and $z(s,k) = z^*(s,k)$ for every commodity. The marginal value of arc e is $MV(e) = T^*(C(e) + 1) - T^*(C(e))$. Code, validation checks and full outputs are in the attached notebook.

Appendix E. Final routes by stream

Source-stream commodities that receive no traffic are omitted. Latency is the sum of link latencies on the path. Shaded rows are latency-sensitive streams.

Stream	Source	Path	Mbps	Latency (ms)
UK-L	S1	S1→A→C→UK	6	27
UK-L	B	B→C→UK	6	17
UK-L	C	C→UK	4	7
UK-T	S1	S1→A→B→C→UK	4	44
UK-T	S1	S1→A→B→D→UK	3	61
UK-T	S2	S2→A→B→D→UK	5.5	63

Stream	Source	Path	Mbps	Latency (ms)
UK-T	S2	S2→A→C→B→D→UK	0.5	66
UK-T	S3	S3→B→A→C→UK	2	40
UK-T	B	B→A→C→UK	4.5	30
UK-T	B	B→D→UK	9.5	34
UK-T	C	C→UK	3.5	7
UK-T	C	C→A→B→D→UK	1.5	57
US-L	S2	S2→A→C→US	5	31
US-L	B	B→C→US	8	19
US-L	C	C→US	5	9
US-T	B	B→A→C→US	7	32
US-T	B	B→D→US	2	32
US-T	C	C→A→B→D→US	6	55

Table E1. Splittable model, final solution (18 routes, 83 Mbps).

Stream	Source	Path	Mbps	Latency (ms)
UK-L	S2	S2→A→C→UK	5	29
UK-L	B	B→C→UK	6	17
UK-L	C	C→UK	1	7
UK-T	S1	S1→A→B→D→UK	4	61
UK-T	S2	S2→A→B→C→UK	6	46
UK-T	S3	S3→B→A→C→UK	7	40
UK-T	B	B→D→UK	14	34
UK-T	C	C→UK	5	7
US-L	S1	S1→A→C→US	5	29
US-L	B	B→D→US	8	32
US-L	C	C→US	5	9
US-T	S1	S1→A→B→C→US	6	46
US-T	S3	S3→B→A→C→US	3	42
US-T	B	B→D→US	2	32
US-T	C	C→US	6	9

Table E2. Single-path model, final solution (15 routes, 83 Mbps).